Introduction to Tuna Stock Assessment

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Cefas

Stock Assessment Process

Data Preparatory Meeting

- Biological information
- Catch and fisheries statistics
- Indices of abundance
- Auxiliary information
- Potential assessment models and options

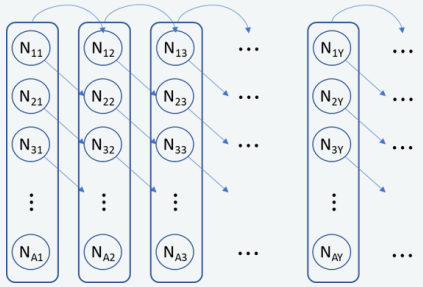
Stock Assessment Meeting

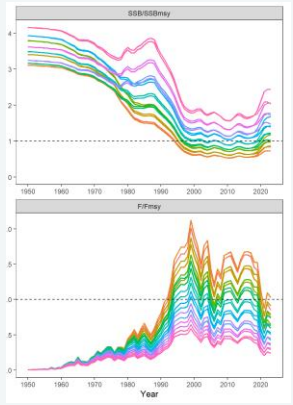
- Run assessment models
- Evaluate model diagnostics
- Determine stock status
- Run projections

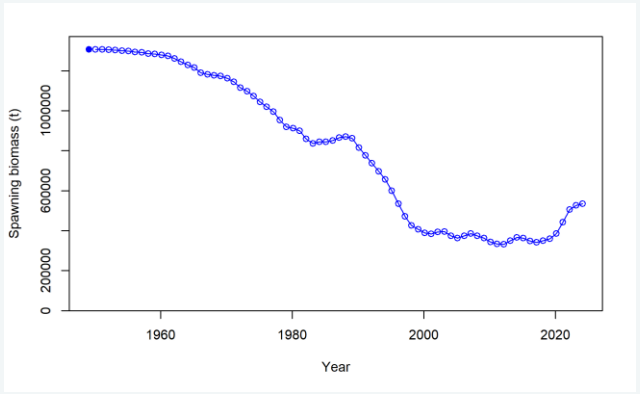
Quantitative estimates of the status of fish stocks and the intensity of fishing upon them

Stock Assessment Process

Age-structured







Model runs

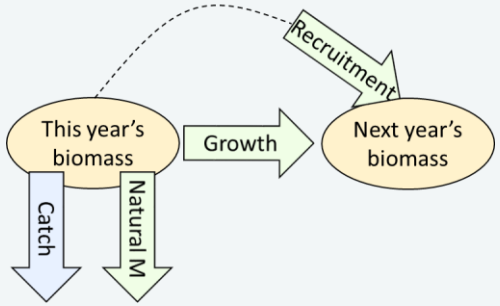
Sensitivity

Diagnostics

Reference case

Uncertainty grid

Surplus production



ss3diags

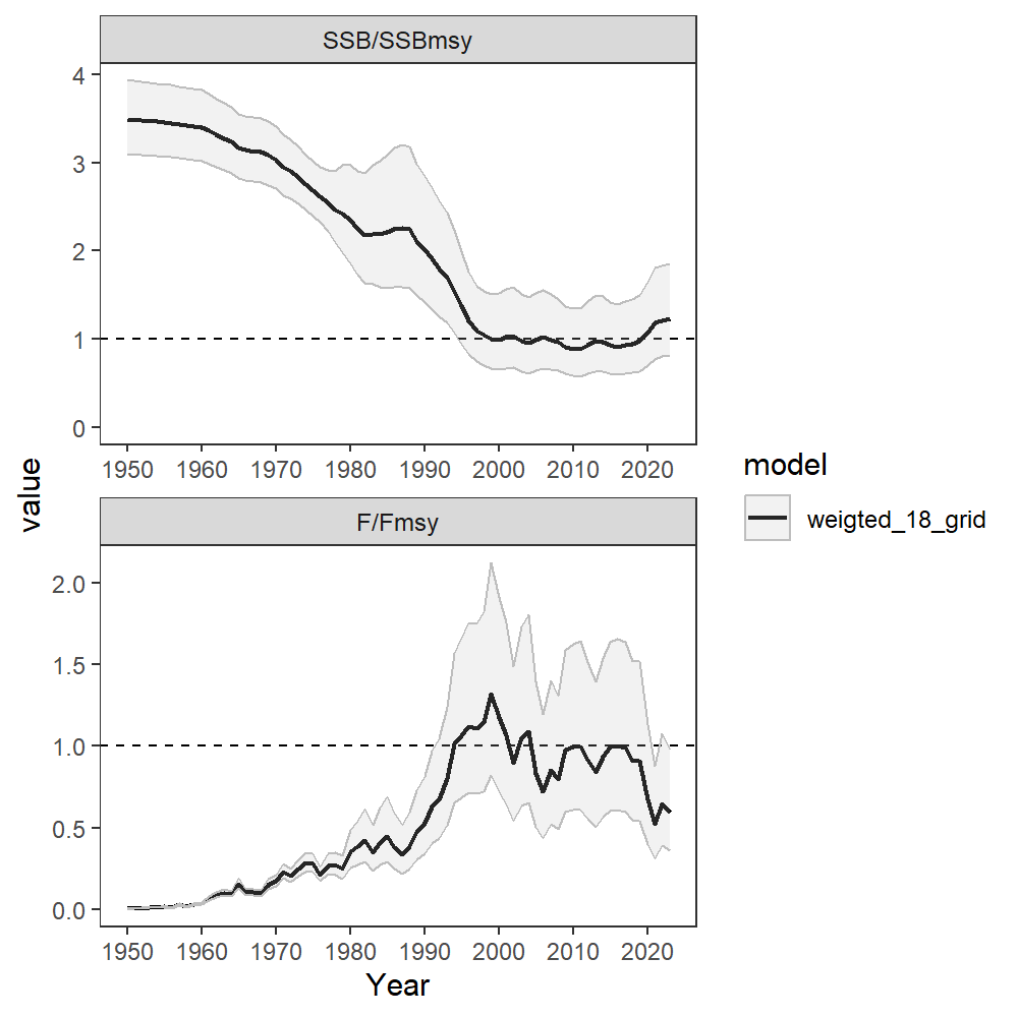
Factor	# Levels	Scenarios
Natural mortality	3	0.26, 0.32, 0.39
Steepness	3	0.7, 0.8, 0.9
Index	2	Acoustic or PS

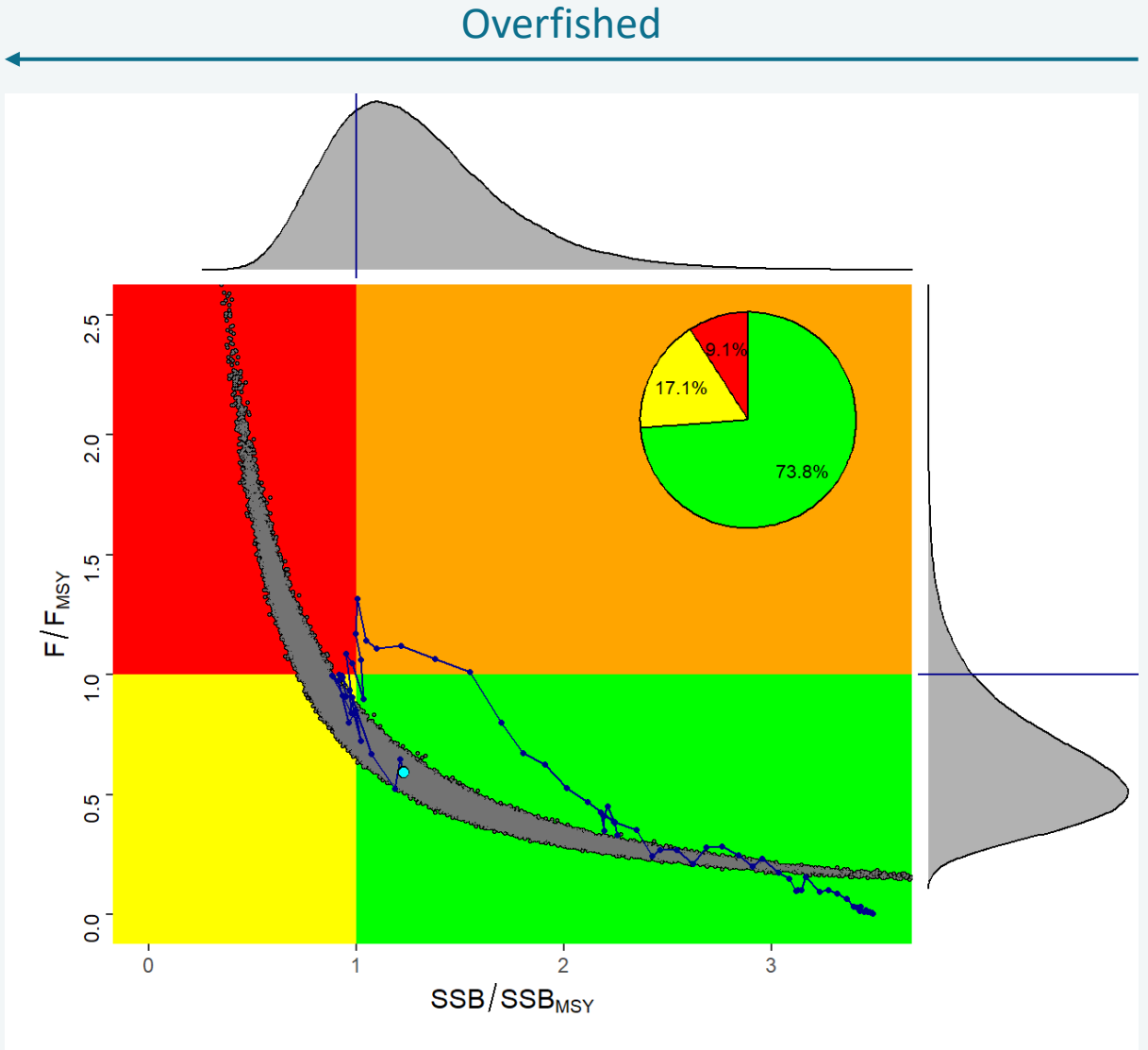
Outputs – Kobe plot

B_{MSY}: Biomass at Maximum Sustainable Yield
F_{MSY}: Fishing mortality corresponding to MSY and B_{MSY}

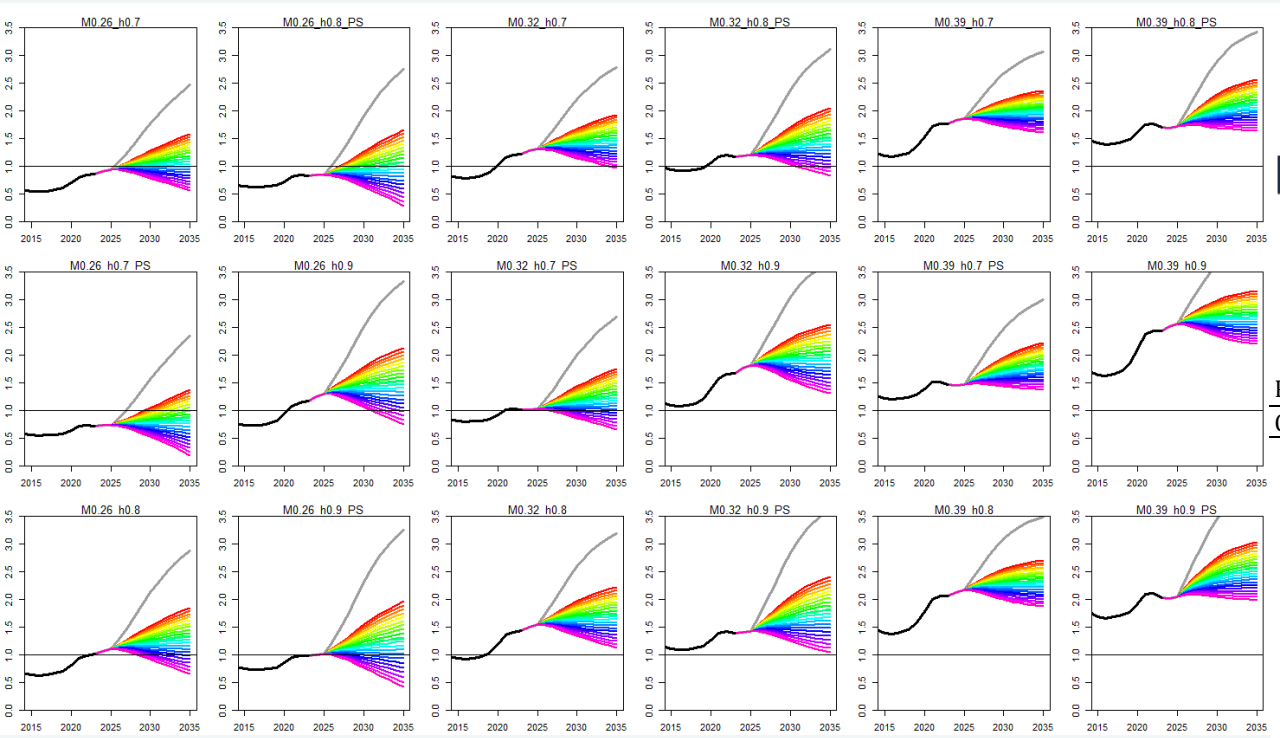
	Stock overfished ($B < B_{MSY}$)	Stock not overfished ($B > B_{MSY}$)
Stock subject to overfishing ($F > F_{MSY}$)	Overfished and overfishing occurring	Not overfished and overfishing occurring
Stock not subject to overfishing $F < F_{MSY}$)	Overfished and overfishing not occurring	Not overfished and overfishing not occurring

Outputs – Trajectories and Kobe plot





Outputs – Projections and Kobe matrix



Probability SSB>=SSB_{MSY}

Catch (t)	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038
50000	86%	91%	94%	97%	98%	99%	100%	100%	100%	100%	100%	100%	100%
52500	86%	90%	94%	96%	98%	99%	100%	100%	100%	100%	100%	100%	100%
55000	85%	90%	93%	96%	97%	99%	99%	100%	100%	100%	100%	100%	100%
57500	85%	89%	92%	95%	97%	98%	99%	99%	100%	100%	100%	100%	100%
60000	85%	88%	91%	94%	96%	97%	98%	99%	99%	100%	100%	100%	100%
62500	84%	88%	90%	93%	95%	96%	97%	98%	99%	99%	99%	100%	100%
65000	84%	87%	89%	92%	94%	95%	96%	97%	98%	98%	99%	99%	99%
67500	83%	86%	89%	90%	92%	94%	95%	96%	97%	97%	98%	98%	99%
70000	83%	85%	87%	89%	91%	92%	93%	94%	95%	96%	96%	97%	97%
72500	83%	85%	86%	87%	89%	90%	91%	92%	92%	93%	94%	95%	95%
73011	83%	85%	86%	87%	88%	89%	90%	91%	92%	93%	93%	94%	95%
75000	83%	84%	85%	86%	87%	88%	88%	89%	89%	90%	91%	92%	92%
77500	82%	83%	84%	84%	85%	85%	85%	86%	86%	87%	87%	88%	88%
80000	82%	83%	83%	83%	83%	83%	83%	83%	83%	83%	83%	83%	83%
82500	82%	82%	82%	81%	81%	80%	80%	80%	79%	79%	78%	78%	78%
85000	81%	81%	80%	79%	79%	78%	77%	76%	75%	74%	73%	73%	72%
87500	81%	81%	79%	78%	76%	75%	73%	72%	71%	70%	68%	67%	66%
90000	81%	80%	78%	76%	74%	72%	70%	68%	66%	65%	63%	62%	60%
92500	80%	79%	77%	74%	71%	69%	67%	64%	62%	60%	58%	56%	55%
95000	80%	78%	75%	72%	69%	66%	63%	60%	58%	56%	54%	52%	51%
97500	80%	77%	74%	70%	67%	63%	60%	57%	54%	52%	50%	49%	48%
100000	79%	77%	73%	68%	64%	60%	56%	53%	50%	48%	47%	46%	45%

Thank you for listening

To find out more contact:

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